AeroSense Bio

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Development of a Bionic Olfactory Edge Computing System Integrating Drosophila Neural Models and Live-Cell Sensor Arrays

Authors
Yi-Chia Fang · Pin-Che Huang · Chun-Ting Chou · Yi-Jung Chen · Ching-Che Chiang

Advisor
Prof. Chung-Chuan Lo
Vice Dean, College of Life Science & Medicine · Computational Neuroscience / NeuroAI

Institution
National Tsing Hua University, Taiwan

01 Abstract

Odor recognition underpins agricultural disease monitoring, environmental sensing, food safety, and medical diagnostics (e.g., lung cancer). Conventional electronic-nose systems still struggle with sensitivity, selectivity, and long-term baseline drift. We propose AeroSense Bio, a bio-digital olfactory platform that (1) reconstructs Drosophila olfactory receptors (ORs) and co-receptor Orco in HEK293T cells with a GCaMP6 calcium reporter, (2) decodes the resulting $\Delta F/F_0$ tensor via a neuromorphic AL→MB→SNN pipeline inspired by the fly olfactory circuit, and (3) deploys the full system in a portable hardware module for in-field and bedside point-of-care testing.

02 Background

Limits of conventional e-noses

Metal-oxide and conducting-polymer arrays suffer baseline drift over long runs and lack specificity for structurally similar odorants — a critical bottleneck for complex VOC mixtures.

Nature's solution — the fly olfactory system

Drosophila detects ppb-level VOCs at minuscule metabolic cost. The antennal lobe (AL) sharpens contrast via lateral inhibition; Kenyon Cells (KC) in the mushroom body implement locality-sensitive hashing, projecting dense signals into a sparse high-dimensional code — a mathematically optimal blueprint for odor classification.

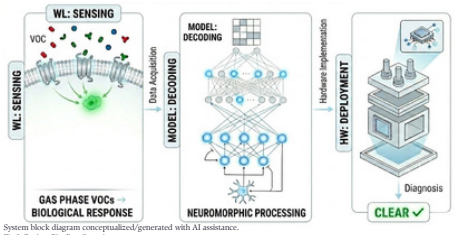
Why living cells?

HEK293T cells offer low background currents, continuous protein turnover (eliminating receptor aging), and a scalable array substrate — combining the specificity of biology with the reproducibility of engineered systems.

ppb-level
sensitivity target across a fly-brain-inspired edge device

03 Pipeline

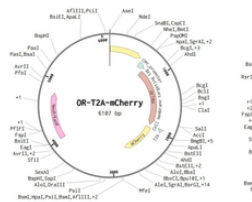
An end-to-end bio-digital pipeline integrating living-cell sensing, neuromorphic decoding, and edge hardware deployment.



System block diagram conceptualized/generated with AI assistance.
Fig 1. Project Pipeline Overview.

04 Bio-Sensor Engineering

Plasmid A



Plasmid B

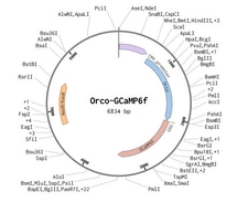
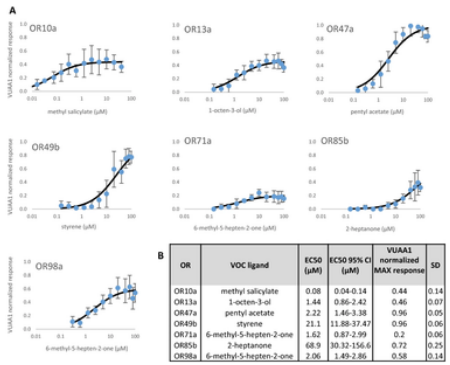


Fig 2. Plasmid Architecture. (Plasmid maps generated with Benchling.)

Host System: Heterologous expression of Drosophila Olfactory Receptors (ORs) and the co-receptor Orco in human HEK293T cells .

Auxiliary Enhancement: Co-expression of Odorant Binding Proteins (OBPs) to improve the solubility and delivery of hydrophobic VOCs in the aqueous culture medium.

Signal Readout: Intracellular Ca2+ influx induced by receptor-odorant binding is converted into real-time fluorescence spikes (Δ F/F0).



B

OR	VOC ligand	EC50 (μM)	EC50 95% CI (μM)	VUAA1 normalized MAX response	SD
OR10a	methyl salicylate	0.08	0.04-0.14	0.44	0.14
OR13a	1-octen-3-ol	1.44	0.86-2.42	0.46	0.07
OR47a	pentyl acetate	2.22	1.46-3.38	0.96	0.05
OR49b	styrene	21.1	11.88-37.47	0.96	0.06
OR71a	6-methyl-5-hepten-2-one	1.62	0.87-2.99	0.2	0.06
OR85b	2-heptanone	68.9	30.32-156.6	0.72	0.25
OR98a	6-methyl-5-hepten-2-one	2.06	1.49-2.86	0.58	0.14

Fig 3. Baseline concentration-dependent response of wild-type ORs to target VOCs (Adapted from Zboray et al., 2023).

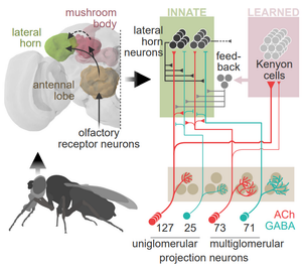
05 AI-Assisted Receptor Optimization

Rational Design: Utilizing AlphaFold 3 to predict the 3D structures of OR-Orco heterodimers .

Molecular Docking: Simulating VOC-receptor interactions in the S1-S4 transmembrane pockets using AutoDock Vina or DiffDock to calculate binding energies.

Directed Evolution: Employing FoldX to predict the energetic effects of mutations (ΔΔG) and designing site-directed mutations (e.g., Ala to Phe substitutions) to enhance affinity for specific biomarkers.

06 Neuromorphic Odor Decoding



Odor signals are transformed from receptor responses (ORNs) through the Antennal Lobe (AL) for contrast enhancement, and further encoded into sparse representations in the Mushroom Body (MB), enabling robust odor discrimination.

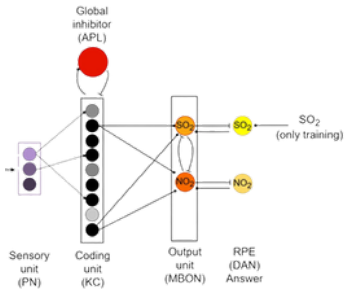


Fig 4. Key advantage Sparse binary encoding separates high-D overlapping odor signatures with a fraction of the compute cost of PCA + SVM — enabling real-time edge inference.

SNN pipeline

① Input representation

ΔF_norm fluorescence change per receptor → fixed low-D feature vector per odor

② AL layer · contrast enhancement

Pattern coding (normalization) + global inhibition (α) + lateral inhibition (β) → removes common background and reduces redundancy

③ MB layer · sparse coding (LSH)

Sparse random projection (5-10 PNs → each KC, 1000-5000 KCs total), Winner-Take-All (top 5%) → high-D binary signature

④ Classifier · edge deployment

Similarity / linear classifier → Spiking NN for ultra-low-power inference on Raspberry Pi

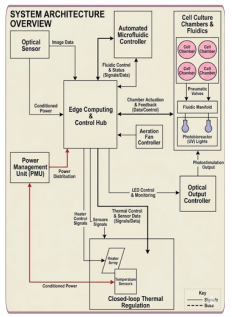
07 Edge-Integrated Apparatus

Modular 4-Layer Chamber: Separates computing, microfluidics, optics, and life-support.

Sterile Fluidics: Septum-piercing interfaces and 0.22μm PTFE filters guarantee contamination-free gas delivery.

Environmental Resilience: Closed-loop thermal control and automated UV-C sterilization maintain long-term cell viability.

Edge Computing Hub: Raspberry Pi and Arduino integration drives real-time neuromorphic inference and hardware automation.



System block diagram conceptualized/generated with AI assistance.
Fig 5. Hardware System Architecture Overview.

Medical POCT: Non-invasive breath diagnostics.
Agriculture: Early crop disease detection.
Environment: Real-time VOC monitoring.

08 Methodology & Target Metrics

Phase 1 POC:
Validation of GCaMP6s response via VUAA1 agonist (50 μM) with a target ΔF/F0 spike > 200%.

Performance Goals:
Accuracy: > 85% for single VOC classification.

Discriminative Power:
> 75% for mixed disease vs. healthy samples.

Robustness:
ROC-AUC > 0.85 validated via 5-fold cross-validation.

09 References

[1] Zboray K., et al. Sci. Rep. 13:21757 (2023). High-throughput ligand profile characterization in novel cell lines expressing seven heterologous insect olfactory receptors for the detection of volatile plant biomarkers.
[2] Butterwick J.A., et al. Nature 560:447–452 (2018). Cryo-EM structure of the insect olfactory receptor Orco.
[3] Bates et al. Current Biology 30, 3183–3199 (2020). Complete Connectomic Reconstruction of Olfactory Projection Neurons in the Fly Brain.

10 Contact Information

AeroSense Bio
aerosensebio@gmail.com
Evangeline (Yi-Chia) Fang, COO
oh180705@gmail.com

Instagram LinkedIn